

Functional genetic report · longevity

Taste and Smell

Sérgio F · Report date: 02-10-2026

How genes shape what they feel in taste and smell, and what this changes in food choices. Olfactory is also a marker of brain aging. Only genes (chip + local imputation + GWAS Catalog scores).

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Summary

19

SNPs analyzed

12

in the chip

7

imputed ($DR2 \geq 0.8$)

0

SNP not found in the data

1

Polygenic scores

The basic flavors are detected by families of receptors in the language: TAS2R (yellow, ~25 genes), TAS1R2/TAS1R3 (sweet), TAS1R1/TAS1R3 (umami), CD36 (fat), sodium channels (salted) and TRPV1 (spicy). The smell uses ~400 olfactory receptors and, for some odors, a single variant changes the perception a lot. These receptors also exist outside the mouth and nose (respiratory pathways, intestine), where they signal bacteria and nutrients. Food preferences (appetite, satiety, taste for food groups) are in the appetite report and eating behavior; here the focus is sensory perception.

In summary: it is an average taster of bitter (TAS2R38 PAV/AVI), has high sensitivity to fat (CD36 GG) and a genetic taste for the above-average sweet (the 82nd percentile). In the smell, it has a copy of the OR7D4 form that makes androstenedione less unpleasant and a copy of the OR6A2 allele that gives the flavor "to the coriander". It is heterozygous in UGT2A1, the locus of olfactory loss in viral infections.

Favourable

- **TAS2R38 PAV/AVI**: medium taster of bitterness; feels the bitter compounds of the crucifers without finding them intense, and the upper respiratory tract receptor is functional (a PAV copy).
- **CD36 rs1761667 GG**: form with more CD36 in the tongue, therefore greater sensitivity to fatty acids (associated with choosing less fat).
- **TAS1R3 CC**: normal sensitivity to sugar.
- **Olfactory receptors without severe loss variants**: OR5A1 and OR2J3 homozygotes for the most common allele.

Unfavourable

- **Genetic taste for sweets at the 82nd percentile** (5 loci score): more will of sweet foods than retatrutide attenuates.
- **UGT2A1/UGT2A2 rs7688383 CT**: a copy of the allele that increases by ~11% the loss of smell and taste in viral infections.
- **OR6A2 CA** (coheses "the soap", slight tendency) and **OR7D4 RT/WM** (androstenedione less perceptible): differences in perception, with no effect on health.

Interpretation: common sensory profile, without severe genetic losses. The most useful information is behavioral: the genetic taste for sweet above average and the normal sensitivity to bitter of crucifers. The smell deserves to be maintained because its loss with age is an early marker of cognitive decline.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 12 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 1 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

There are no scores published in the PGS Catalog for taste or smell. The sweet food taste score uses 5 loci from GWAS Catalog (a study of food preferences) and is at the 82nd percentile. We excluded loci from studies of other populations. The olfactory and coriander scores had less than 3 loci with usable effect size and were not used.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
I like sweet foods (GWAS Catalog score, 5 loci)	C_DOCE	5 of 5 (100%)	p82	Typical / no judgement

Bitter (TAS2R38, TAS2R19/TAS2R14, gustina)

TAS2R38 detects phenylthiocarbamide and PROP (and the glucosinolates of the crucifers). Three SNPs define the haplotype PAV (proofer) or AVI (non-tester). The TAS2R14/TAS2R19 cluster in the 12 chromosome controls the bitterness of quinine and caffeine.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
TAS2R38 A49P	rs713598	CG	chip	—	—	—	Receptor of the PTC/PROP bitterness and the compounds of the crucifers; with rs1726866 and rs10246939 it defines the haplotype PAV (proofer) or AVI (non-tester). (strong evidence)	Typical / no judgement
TAS2R38 A262V	rs1726866	GA	chip	—	—	—	According to SNP of the haplotype TAS2R38. (strong evidence)	Typical / no judgement
TAS2R38 V296I	rs10246939	TC	chip	—	—	—	Third SNP of the TAS2R38 haplotype; the receptor also exists in the upper respiratory tract, where it detects bacterial signaling molecules. (strong evidence)	Typical / no judgement
TAS2R19 R299C	rs10772420	AA	chip	—	—	—	Bitter locus of quinine and grapefruit (TAS2R19/TAS2R14/PRH1 region). (strong evidence)	Typical / no judgement
TAS2R14/ PRH1	rs2597979	CC	imputation DR2=0.98	—	—	—	Second sign of quinine and caffeine bitterness in the 12 chromosome cluster. (moderate evidence)	Typical / no judgement
CA6 Ser90Gly	rs2274333	AG	imputation DR2=0.86	—	—	—	Gastin (carbonic anhydrase 6 of saliva), linked to zinc and the growth of taste buds; G allele was associated with lower sensitivity to PROP. (weak evidence)	Typical / no judgement

Favourable

- TAS2R38 heterozygous in the 3 SNPs (PAV/AVI, medium taster)
- TAS2R19 rs10772420 AA and TAS2R14 rs2597979 CC (homozygous for common alleles)

Unfavourable

- CA6 (gustin) Ser90Gly AG: a copy associated with less sensitivity to PROP

Interpretation: Medium taster: feels the crucifers as slightly bitter. In the superprovers (PAV/PAV) the bitter moves further away from the vegetables; in its case the effect is small. The TAS2R38 of the upper respiratory tract reacts to the signs of some bacteria and helps the defense of the nose and sinuses; an PAV copy gives intermediate response.

Sweet, fat, salty and hot

The sweet is detected by the dimer TAS1R2/TAS1R3, the fat by CD36 and GPR120, the salty by the epithelial sodium channel (SCNN1) and the hot by TRPV1.

Polygenic scores of this theme: I like sweet foods (GWAS Catalog score, 5 loci) (C_DICE): the 82nd percentile
Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
TAS1R3 -1572C>T	rs35744813	CC	chip	—	—	—	Sweet (and umami) receiver: the promoter's T allele reduces sucrose sensitivity. (moderate evidence)	Typical / no judgement
TAS1R2 Ile191Val	rs35874116	CC	chip	—	—	—	Sweet receiver: the C allele (Val191) was associated with lower sugar consumption. (weak evidence)	Typical / no judgement
CD36	rs1761667	GG	chip	—	—	—	CD36, the fat flavor receptor in the tongue: A allele reduces CD36 expression and sensitivity to fatty acids. (moderate evidence)	Typical / no judgement
SCNN1B	rs239345	TA	chip	—	—	—	An epithelial sodium channel (salt flavor): allele A was associated with other salt sensitivity. (weak evidence)	Typical / no judgement
TRPV1 Ile585Val	rs8065080	CC	chip	—	—	—	Capsaicin receptor (pinning) and heat; also modulates salt perception. (weak evidence)	Typical / no judgement

Favourable

- CD36 GG (more sensitive to fat)
- TAS1R3 -1572 CC (normal sensitivity to sucrose)
- TAS1R2 Val191/Val191 (associated with less sugar consumption)

Unfavourable

- Genetic taste for sweets at the 82nd percentile

Interpretation: Sweet receptors are normal; the sweetness score measures preference (reward circuits), not the sensitivity of the tongue, and is consistent with the sweetness score of the appetite report. Retatrutide reduces the reward associated with sweet and fat foods. SCNN1B TA and TRPV1 CC: common variants without practical reading.

Olfactory receivers with known effect

For some odors, a single receptor explains almost all the difference between people: androstenone (OR7D4), β -ionone (OR5A1), coriander aldehydes (OR6A2), cut grass smell (OR2J3).

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
OR7D4 R88W	rs61729907	GA	chip	—	—	—	Androstenone receptor (almiscarate/ urine odor); R88W and T133M (WM form) reduce sensitivity and make the odor less unpleasant. (strong evidence)	Typical / no judgement
OR7D4 T133M	rs5020278	GA	imputation DR2=1.00	—	—	—	Second SNP of the form WM of OR7D4. (strong evidence)	Typical / no judgement
OR5A1 N183D	rs6591536	AA	chip	—	—	—	Receptor of β -ionone (floral/violet aroma of fruits and wine): this SNP explains most of the differences in sensitivity to this odor. (strong evidence)	Typical / no judgement
OR6A2	rs72921001	CA	chip	—	—	—	Allele A is associated with the feeling of coriander flavored with soap. (moderate evidence)	Typical / no judgement
OR2J3 T113A	rs28757581	AA	imputation DR2=1.00	—	—	—	Cis-3-hexen-1-ol receiver (thick grass). (moderate evidence)	Typical / no judgement

Favourable

- OR5A1 rs6591536 AA and OR2J3 rs28757581 AA (common alleles)

Unfavourable

- OR7D4 R88W and T133M heterozygous (one WM copy)
- OR6A2 rs72921001 CA (a copy of the "soap" allele)

Interpretation: Differences in perception without effect on health. With an WM copy of OR7D4, androstenone (present in male pork and sweat) is less intense and less unpleasant.

Loss of smell with age and after infections

The loss of smell with age is frequent (about half of people after the 65 years) and is one of the earliest signs of Parkinson's disease and Alzheimer's; anosmia is also associated with more mortality in cohort studies. Viral infections (such as COVID-19) are another frequent cause.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
OR (11q11)	rs11228623	TC	imputation DR2=1.00	—	—	—	Cluster of olfactory receptors in the 11 chromosome: locus of olfactory dysfunction in older adults. (moderate evidence)	Typical / no judgement
OR (11q11)	rs12786376	AC	imputation DR2=1.00	—	—	—	Second sign of the same cluster, associated with olfactory dysfunction. (moderate evidence)	Typical / no judgement
UGT2A1/UGT2A2	rs7688383	CT	imputation DR2=0.98	T	1	0.39	Enzymes that eliminate odorants in the olfactory epithelium; T allele increases by ~11% the loss of smell/palade in viral infection (COVID-19). (moderate evidence)	Typical / no judgement

Favourable

- No homozygosity for risk alleles

Unfavourable

- UGT2A1/UGT2A2 rs7688383 CT
- OR Cluster of the 11 chromosome: heterozygous in both SNPs (rs11228623 TC, rs12786376 AC)

Interpretation: Average genetic risk of smell loss. The enzymes UGT2A1/UGT2A2 of the olfactory epithelium eliminate the odorants; the T allele slightly increases the probability of losing the smell in a viral infection.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Intercellular Communication	CA6 Ser90Gly AG	TAS2R38 PAV/AVI; common olfactory receptors	Common sensory profile.
Metabolic Homeostasis Decline	I like p82 sweets	CD36 GG; TAS1R2 Val/Val	Mixed; compensated by the food plan and the retatrutide.
Neuroinflammation	UGT2A1 CT; cluster OR heterozygous	—	Olfactory is an early marker of neurodegeneration.
Chronic Inflammation	—	Functional TAS2R38 (1 copy PAV) in the respiratory tract	Innate defense of the middle nose.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Olfactory training	Smelt 4 different odors (e.g. pink, lemon, clove, eucalypt) 20 seconds each, twice a day; is the treatment with more evidence in the loss of smell and maintains the stimulated olfactory system. <i>suggestion (not applied)</i>	UGT2A1 CT; brain marker
Protecting the olfactory epithelium	Avoiding nasal sprays or gels with zinc (can cause permanent anosmia) and exposure to solvents without ventilation. <i>suggestion (not applied)</i>	Loss of smell
Crucifers despite the bitter	Keeping broccoli, kale, rocket and cress; the bitter is moderate for a medium taster, and cooking little to steam or adding mustard powder keeps the sulforaphane. <i>suggestion (not applied)</i>	TAS2R38 PAV/AVI
Sweet food	Do not have candy at home and use whole fruit for the sweet; the genetic taste for the sweet is above average. <i>suggestion (not applied)</i>	Sweet p82 score
Drugs and flavor	If metallic or altered taste appears, remember that lithium, telmisartan and isotretinoin (dry mouth) can change the taste; talk to the doctor before changing any dose. <i>suggestion (not applied)</i>	Drugs in use
Zinc and B12	Zinc and B12 sufficient for food (marish, meat, legumes): the lack of anyone reduces taste. <i>suggestion (not applied)</i>	Gustina (CA6) AG
Coordination with the plan	Without alcohol. D3 10 000 IU and omega-3 3.2-4 g/day (already taking fish oil). Melatonin 0.3-1 mg. Curcumin, luteolin, apigenin and quercetin in pause while ALT is high. <i>suggestion (not applied)</i>	V2 plan rules

Limitations

- Most of the ~400 olfactory receptors have variants whose effect is unknown; only some odors have a known main gene.
- Olfactory GWAS and olfactory loss are small; there is no published polygenic score.
- The taste score for sweets measures declared preference, not the sensitivity of the language.
- The percentiles compare with the reference population (n=633). Only genes: does not use smell or taste tests.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population (n=633).
- Scores: Simple GWAS Catalog score (local copy) for sweet food taste (5 loci).
- Main references: Kim 2003 (TAS2R38 and PTC); Lee 2012 (TAS2R38 in upper respiratory tracts); Reed 2010 (cluster TAS2R of chromosome 12); Hwang 2018 (GWAS of bitter perception); Fushan 2009 (TAS1R3 and sucrose sensitivity); Keller 2007 (OR7D4 and androstenone); Jaeger 2013 (OR5A1 and β -ionone); Eriksson 2012 (OR6A2 and coriander); Shelton 2022 (UGT2A1/UGT2A2 and loss of smell in COVID-19); May-Wilson 2022 (GWAS of food preferences); Hummel 2009 (Olfactive Training).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).